

Collection system -

- The Aligarh city handles municipal solid waste management based on the municipal area. Currently municipal city area is divided into 5 zones and these sanitary zones are further divided into wards to facilitate the collection and transportation of solid waste.
- Most of the solid waste generated is contributed from household that is about 60% of waste is domestic waste. For Primary collection, door to door collection of MSW is done by deploying Mini Tipper and Tricycle Rickshaws.
- For door to door collection, stationary container system should be adopted to cover max. households on daily basis.
- For rest commercial, institutional, ~~and~~ market, etc. waste big size of waste containers are proposed as community bins. Therefore haul container system with mechanical lifting will be adequate to follow and will require lesser manpower.
- For given manually loaded container in Door to Door collection of waste, assume the following data.
- $H = 8$ hrs work day $N_d = 3$ trips/d. $W = 15\%$
 $t_1 = 15$ mins $t_2 = 20$ mins. $dbc = 3$ min $S = 0.25$ hr
 $a = 0.016$ $b = 0.011$ Speed limit = 40 km/hr
- container utilization factor, $f = 0.8$
 one way haul distance = 20 km
 $C_n = \text{avg. 2 containers per location}$
 $V_c = 10$ $\rho = 177 \text{ kg/m}^3$
 Collection frequency - Every day
 Total waste to be collected in Overall 5 zones = 1867 mt/day

$$H = \frac{(t_1 + t_2) + N_d T_c}{1 - W}$$

$$\therefore T_{scs} = \frac{H(1-W) - (t_1 + t_2)}{N_d} = \frac{8(1-0.15) - (15+20)/60}{3}$$

$$= 2.07 \text{ hr/trip}$$

Now, pick up time, $P_{scs} = T_{scs} - a - bx - S$

$$= 2.07 - 0.016 - 0.011(20) - 0.25$$

$$= 1.584 \text{ hr/trip}$$

⇒ No. of pickup locations served per trip, $N_p = \frac{60 \times P_{scs} \times n}{t_p}$

$n = \text{no. of collectors}$
 $= 2$

$t_p \rightarrow \text{pick up time per location}$
 $= \text{dbc} + 0.18(\text{km}) + 0.014(\text{PRH})$

$$N_p = \frac{60 \times 1.584 \times 2}{2 + 0.18(20) + 0.014(0)}$$

$$N_p = 80.6 \approx 81 \text{ locations/trip}$$

Volume of waste generated per pickup location per ~~trip~~ day, assuming avg. population divided in each sub zone for collection is same throughout.

$$\text{Waste} = \frac{186.7 \times 1000}{5 \times 18} = 20745 \text{ Kg/day/subzone}$$

now density of MSW of Aligarh city = 177 Kg/m^3

$$\text{Volume of waste per day per pickup location} = \frac{20745}{177} \times \frac{1}{81}$$

$$V_p = 1.447 \text{ m}^3/\text{location/d}$$

⇒ Volume of waste to be collected is 1.447 m^3 per location per day as collection will be on daily basis.

$$\text{Vol}^{\sim} \text{ of container of vehicle} = \frac{V_p \times n_p}{\lambda} \quad (\text{Compaction Ratio } \lambda = 2.5)$$

$$= \frac{1.447 \times 81}{2.5} = 46.88$$

$$\approx 47 \text{ m}^3$$

Based on volume of ~~truck~~ vehicle, we need to have more number of collection in each subzone as truck volume with this much

volume will have difficulties entering city area.

Assume there are 3^{times} more collections in each subzone volume of container will be nearly 16 m^3 . We can also consider junction for a subzone which can act as transfer station through which large amount of waste can be collected through large volume vehicles.

$$\text{volume of container} = 16 \text{ m}^3$$

$$\rightarrow \text{No. of vehicles} = 3 \times 5 \times 10 = 270 \text{ for door to door collection based on 2055 data.}$$

→ For collection of waste from commercial spaces in design of haul container system, take the same data and
 $\text{dbc} = 20 \text{ min}$ $x = 30 \text{ km}$ pickup time & unloading time = 0.4 hr/trip

$$\begin{aligned} T_{\text{hes}} &= P_c + U_c + \text{dbc} + a + bx + 5 \\ &= 0.4 + \frac{20}{60} + 0.016 + 0.011(30) + 0.25 \\ &= 1.329 \text{ hrs} \end{aligned}$$

$$\text{No. of trips per day, } N_d = \frac{(1-W)H - (t_1 + t_2)}{T_{\text{hes}}}$$

$$N_d = \frac{(1-0.15)8 - (15+20)/60}{1.329}$$

$$N_d = 4.68 \sim 5 \text{ trips/day}$$

$$\begin{aligned} \text{Actually length of the weekday, } H &= \frac{N_d T_{\text{hes}} + (t_1 + t_2)}{(1-W)} \\ &= \frac{5 \times 1.329 + (15+20)/60}{0.85} \\ &= 8.3 \text{ hrs} \approx 8 \text{ hrs} \end{aligned}$$

→ Therefore, based on the estimated waste generation, the collection of waste would be managed based on the different zones, sub-zones and wards. Frequency of the waste would also be increased for collection management as shown above.